

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems

/ 5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Number : CO3_AT2

Assessment Tool Weightage : 35%

Assessment Tool : Design Task

Duration : 180 Minutes

Marks : 100 Marks

Type of Assessment Conduction : At Home

Evaluation Type : GitHub

Result Generation : Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.

Topic Covered :

Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills. This question paper consists of a total of 40 questions, out of which each student is required to answer any 2 questions. Each question carries 50 marks, and the paper carries a total of 100 marks

QUESTIONS: 2 (2 X 50 = 100) Total Marks: 100 Duration: 3hrs

15. Design a wireless service framework integrating virtual network functions. (BL6)
34. Develop an energy monitoring communication model for public infrastructure

Code of Conduct:

I Yogeswaran/192412615 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided

Communicat io n	15%	Communicate s feedback clearly, professionall y, and engages actively in discussion	Communicate s well with minor issues	Communicatio n lacks clarity or engagement	Poor communica tio n and minimal participation
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15. Design a Wireless Service Framework Integrating Virtual Network Functions

AIM:

To design and simulate a wireless service framework integrating Virtual Network Functions (VNFs) and analyze the total processing delay experienced by data packets in a virtualized network environment

PROGRAM CODE:

```
% Wireless Service Framework using Virtual Network Functions (VNF)
clc;
clear;
close all;

% Number of packets
packets = 1:100;

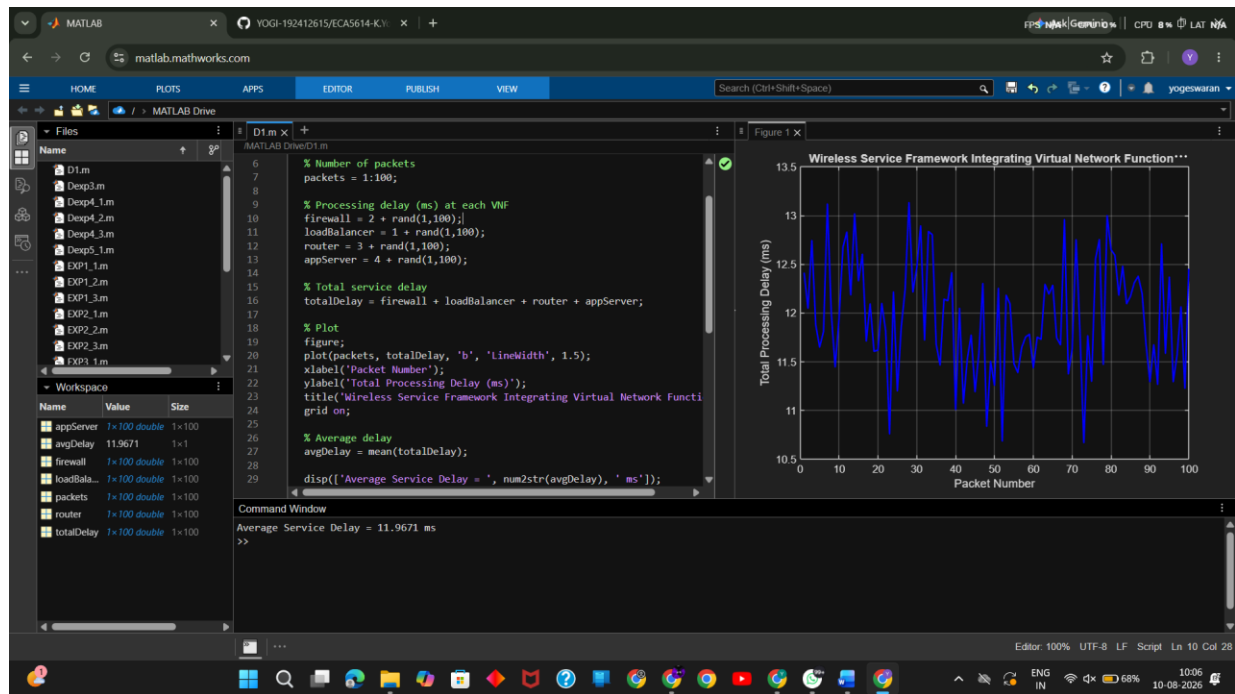
% Processing delay (ms) at each VNF
firewall = 2 + rand(1,100);
loadBalancer = 1 + rand(1,100);
router = 3 + rand(1,100);
appServer = 4 + rand(1,100);

% Total service delay
totalDelay = firewall + loadBalancer + router + appServer;

% Plot
figure;
plot(packets, totalDelay, 'b', 'LineWidth', 1.5);
xlabel('Packet Number');
ylabel('Total Processing Delay (ms)');
title('Wireless Service Framework Integrating Virtual Network Functions');
grid on;

% Average delay
avgDelay = mean(totalDelay);
disp(['Average Service Delay = ', num2str(avgDelay), ' ms']);
```

OUTPUT



RESULT

The wireless service framework integrating Virtual Network Functions was successfully simulated in MATLAB. The graph of packet number versus total processing delay was obtained, and the average service delay was calculated. The simulation shows that virtualization enables flexible service chaining while allowing the processing delay of the network functions to be analyzed effectively.

34. Develop an energy monitoring communication model for public infrastructure

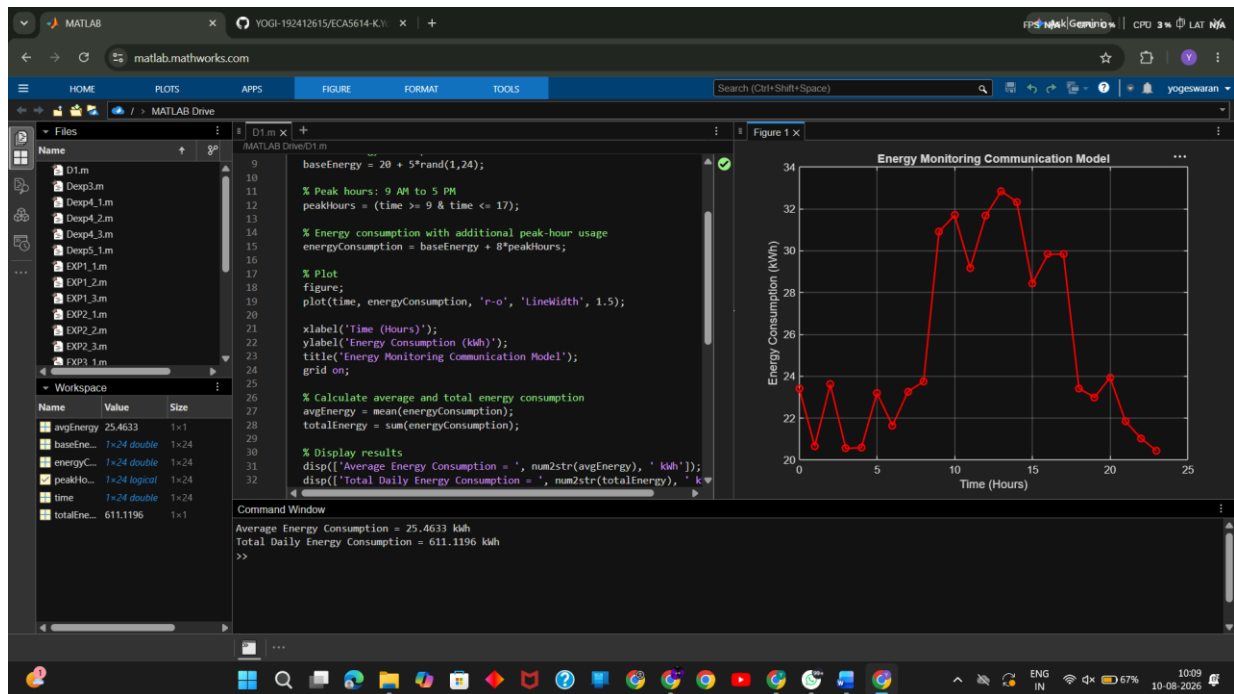
AIM:

To develop and simulate an energy monitoring communication model for public infrastructure and analyze the variation in energy consumption over a 24-hour period.

PROGRAM CODE:

```
clc;
clear;
close all;
% Time: 24 hours
time = 0:23;
% Base energy consumption
baseEnergy = 20 + 5*rand(1,24);
% Peak hours: 9 AM to 5 PM
peakHours = (time >= 9 & time <= 17);
% Energy consumption with additional peak-hour usage
energyConsumption = baseEnergy + 8*peakHours;
% Plot
figure;
plot(time, energyConsumption, 'r-o', 'LineWidth', 1.5);
xlabel('Time (Hours)');
ylabel('Energy Consumption (kWh)');
title('Energy Monitoring Communication Model');
grid on;
% Calculate average and total energy consumption
avgEnergy = mean(energyConsumption);
totalEnergy = sum(energyConsumption);
% Display results
disp(['Average Energy Consumption = ', num2str(avgEnergy), ' kWh']);
disp(['Total Daily Energy Consumption = ', num2str(totalEnergy), ' kWh']);
```

OUTPUT:



RESULT:

The energy monitoring communication model for public infrastructure was successfully simulated in MATLAB. The graph shows the variation in energy consumption over a 24-hour period, with higher consumption during peak hours. The average and total daily energy consumption were calculated, demonstrating the usefulness of wireless energy monitoring for efficient infrastructure management.